# Juan Cruz Dominguez Pistoia

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GitHub: Cruplusplus • Portfolio • LinkedIn

**National Technological University (UTN – FRGP)** General Pacheco, Buenos Aires

Associate Degree in Computer Programming (In Progress - 2nd Year) Expected Graduation: June 2027

Relevant Coursework: Object-Oriented Programming (C#/C++), Data Structures & Algorithms, Relational & Non-Relational Database Systems, Layered Software Architecture, Systems Analysis.

**First Certificate in English (FCE) - Cambridge University Press & Assessment**

B2 First - Independent User Pilar, December 2024

**Instituto Marcelo Torcuato de Alvear** Don Torcuato, Buenos Aires

High School Diploma & Bachelor's in Economics and Business Administration December 2024

## Projects

**Gym Management Web Application (.NET / C#) – Academic Project**

• Built a full-stack gym management web application applying a decoupled N-Tier layered architecture (Presentation, Business Logic, and Data Access layers).

• Designed and normalized a relational database in Microsoft SQL Server, optimizing queries, stored procedures, and data integrity.

• Implemented core business rules on the backend and created integrated client-server form validations.

• Utilized HTML, CSS, JavaScript, and ASP.NET controls to provide a responsive and user-friendly interface.

**Asynchronous Automation Service & Bot (Python) – Personal Project**

• Developed an automated asynchronous background bot utilizing Python, discord.py, and aiohttp.

• Integrated and consumed external RESTful APIs using asynchronous HTTP requests with rate limiting and structured error handling.

• Created a persistent data storage and state tracking system through JSON parsing and file serialization.

**2D Extensible Game Engine & RPG Simulator (C++) – Academic Project**

Repository: github.com/Cruplusplus/TP-Programacion-2.-Extendible-map-RPG-in-C-

• Recreated core 2D RPG engine mechanics implementing advanced Object-Oriented Programming (OOP) concepts, including inheritance and polymorphism.

• Utilized the SFML multimedia library for 2D rendering, input handling, and resource pipeline management.

• Debugged runtime memory leaks and optimized pointer operations and dynamic memory structures.

• Formulated a dynamic loading and parsing subsystem for entities, game objects, and map tiles using JSON data files.

## Skills & Interests

**Technical Skills:**

• Proficient in: C#, .NET / ASP.NET, Microsoft SQL Server (T-SQL, Stored Procedures), C++, Git & GitHub.

• Advanced proficiency in: Python, JavaScript (ES6+), HTML5, CSS3, JSON serialization, REST APIs.

• Skilled in: MongoDB, TailwindCSS, Docker, VS Code, Visual Studio, Compass, SFML.

**Languages:**

• English: Upper-Intermediate / Fluent technical and conversational English (B2 First Certificate - Cambridge).

• Spanish: Native proficiency.

**Interests:**

• Backend architecture, algorithm design, software optimization, and open-source game development.